1. $\int xe^{-x}\,dx$

$u = x \qquad du = dx$

$dv = e^{-x}\,dx \qquad v = \int e^{-x}\,dx = -e^{-x}$

จาก $\int u\,dv = uv - \int v\,du$

$$\int xe^{-x}\,dx = -xe^{-x} - \int -e^{-x}\,dx$$

$$= -xe^{-x} - \int e^{-x}\,dx$$

$$= -xe^{-x} - e^{-x}$$

$$= \frac{-x}{e^{x}} - \frac{1}{e^{x}}$$

$$= \frac{-x+1}{e^{x}}$$

2. $\int x\cos 5x\,dx$

$u = x \qquad du = dx$

$dv = \cos 5x\,dx \qquad v = \int \cos 5x\,dx = \frac{1}{5}\sin 5x$

จาก $\int u\,dv = uv - \int v\,du$

$$\int x\cos 5x\,dx = \frac{x}{5}\sin 5x - \int \frac{1}{5}\sin 5x\,dx$$

$$= \frac{x}{5}\sin 5x - \frac{1}{5}\int \sin 5x\,dx \quad \frac{1}{5}\int \sin \quad x \quad d$$

$$= \frac{x}{5}\sin 5x - \frac{1}{5}\left(-\frac{1}{5}\cos(5x)\right)$$

$$= \frac{x}{5}\sin 5x + \frac{1}{25}\cos 5x + c$$

$$3. \int \cos^{\overset{n}{3}} x \sin^{\overset{m}{2}} x \, dx = \int \cos^{\cancel{3}\,2} x \sin^2 x \, \frac{d(\sin x)}{\cancel{\cos x}}$$

$$= \int \cos^2 x \sin^2 x \, d(\sin x)$$

$$= \int (1-\sin^2 x) \sin^2 x \, d(\sin x)$$

$$= \int (\sin^2 x + \sin^4 x) \, d(\sin x)$$

$$= \frac{\sin^3 x}{3} + \frac{\sin^5 x}{5} + C$$

$$4. \int \sec^2 x \tan^3 x \, dx = \int \cancel{\sec^2 x} \tan^3 x \, \frac{d(\tan x)}{\cancel{\sec^2 x}}$$

$$= \int \tan^3 x \, d(\tan x)$$

$$= \frac{\tan^4 x}{4} + C$$

$$5. \int \frac{\sqrt{16 - x^2}}{x^2} = \int \frac{\sqrt{16}}{x^2} dx - \int \frac{\sqrt{x^2}}{x^2} dx$$

$$= \int \frac{4}{x^2} dx - \int \frac{x}{x^2} dx$$

$$= 4 \int \frac{1}{x^2} dx - \int \frac{x}{x^2} dx$$

$$4 \ln|x^2| - \int \frac{\cancel{x}^{\,1}}{x^2} \frac{d(x^2)}{2\cancel{x}}$$

$$4 \ln|x^2| - \frac{1}{2} \int \frac{1}{x^2} d(x^2 \qquad = 4 \ln|x^2| - \frac{1}{2} \ln|x^2|$$